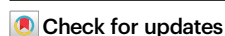


Highly porous metal-organic framework glass design and application for gas separation membranes

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Crystalline metal-organic frameworks (MOFs) exhibit enormous potential application in gas separation, thanks to their highly porous structures and precise pore size distributions. Nevertheless, the inherent limitations in mechanical stability of crystalline MOFs cause challenges in processing MOF powders into bulky structures, particularly for membrane filtrations. Melt-quenched MOF glasses boast excellent processability due to liquid-like properties. However, the melting process diminishes the inherent porosity, leading to reduced gas adsorption capacities and lower gas diffusion coefficients. In this work, we demonstrated that enhancing the porosity of MOF glasses is achievable through topological engineering on the crystalline precursors. Crystalline zeolitic imidazolate frameworks (ZIFs) with large 12-membered rings pores, including AFI and CAN topology, were synthesized by using both structure-directing agents and mixed organic ligands. The large pores are partially preserved in the melt-quenched glass as evidenced by high-pressure CO₂ absorption at 3000 kPa. The $a_g\text{AFI}[\text{Zn}(\text{Im})_{1.68}(\text{blm})_{0.32}]$ glass was then fabricated into self-supported membranes, which shows high gas separation performance, for example, CO₂ permeance of 3.7×10^4 GPU with a CO₂/N₂ selectivity of 14.8.

Excessive energy consumption, particularly from the continuous burning of fossil fuels, has significantly worsened climate change^{1–4}. Extensive research has focused on clean energy production and carbon capture to reduce the release of greenhouse gases, mainly CO₂ emissions, into the atmosphere during the process of extracting, producing, and using energy^{2–4}. MOFs are cage-like structures containing metal nodes and organic ligands connected in infinite arrays⁵.

The highly porous structures and precise pore size distributions of crystalline MOFs give them potential applications in various energy and environment-related gas separations, such as natural gas purification and CO₂ capture^{6–11}. Crystalline MOFs typically require processing, such as pelletization, coating, or sintering, to form bulky structures for device applications, especially for membrane modules^{12,13}. However, their practical implementation is still hampered

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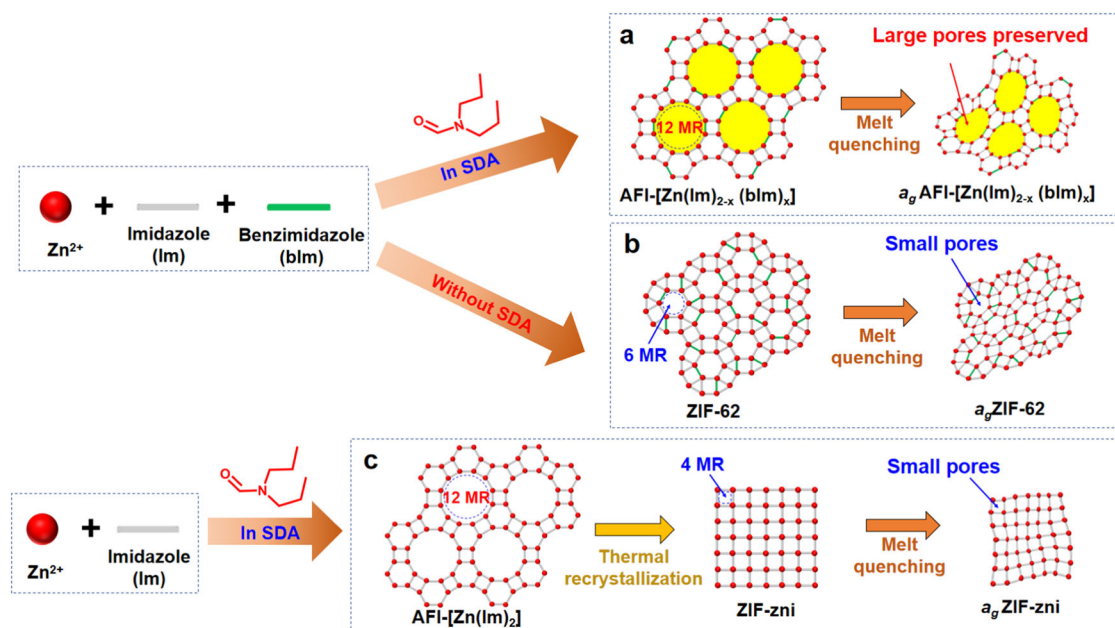


Fig. 1 | Synthesis scheme of meltable MOFs with highly porous topologies by using both bulky amide solvents as SDAs and mixed ligands. SDAs create highly porous topologies without introducing extra ligands. The mixed ligands of Im and blm suppress the recrystallization to dense ZIF-zni phase during the melting process, and provide spatial support for porosity preservation. **a** Synthesis of large-

pore AFI-[Zn(Im)_{2-x}(blm)_x] and *a_g*-AFI-[Zn(Im)_{2-x}(blm)_x] by using both SDAs and mixed ligands. **b** Synthesis of small-pore ZIF-62 (CCDC code SIWJAM) and *a_g*-ZIF-62 by using only mixed ligands. **c** Synthesis of AFI-[Zn(Im)₂] (CCDC code PAJROK), thermal recrystallization ZIF-zni (CCDC code IMIDZB), and *a_g*-AFI-[Zn(Im)₂] by using only SDAs.

by inherent limitations in mechanical stability of microcrystalline powders^{14,15}.

The discovery of MOF glasses, recognised as a fourth category of glass materials, given their dissimilarity to existing inorganic (non-metallic), organic and metallic families, opens up possible opportunities to process MOFs into macroscale, grain-boundary free morphologies¹⁶. ZIFs are a class of MOFs containing tetrahedral metal ions linked by imidazolate ions. A few types of ZIFs have been discovered with the capability to form glasses via melt-quenching¹⁷. These melt-quenched MOF glasses possess similar structures to amorphous silica and display viscoelastic states at temperatures above their glass transition temperatures (T_g). Several forming methods such as hot-pressing and remelting have been explored to fabricate bulk samples of both MOF glasses and MOF crystal glass composites (CGCs)^{18–20}. However, upon glass formation, the inherent porosity in their crystalline precursors is reduced, which would be unfavourable for possible applications in gas separation compared with other pure polycrystalline MOFs^{10,21–25}. Hence, a substantial enhancement in porosity is imperative for MOF glasses to emerge as practical solutions for gas separation challenges. The introduction of pore-inducing agents can enhance the porosity of MOF glass through the decomposition of chemicals added during glass formation, such as polymers^{21,26}. Nevertheless, achieving precise manipulation of porosity in MOF glass at the Ångström scale remains a significant challenge.

Chemical composition of the crystalline MOFs, i.e., the type of metal nodes and organic ligands, is a decisive factor in determining their melting point and decomposition temperature^{27,28}. Thus, to make both meltable and highly porous crystalline MOFs, it is logical to engineer a more porous topology by using similar metal nodes and organic ligands to those present in reported meltable MOFs, e.g., ZIF-4, [Zn(Im)₂, Im=imidazolate], and ZIF-62, [Zn(Im)_{1.75}(blm)_{0.25}, blm=benzimidazolate]. The commonly employed ligand-directed strategy, which utilizes large-sized ligands to enhance porosity in crystalline MOFs, is not readily applicable to meltable MOFs. This is due to the introduction of extra ligands which may potentially lower the decomposition temperature of MOFs²¹.

Large solvent molecules have previously been used as the structure-directing agents (SDAs) to synthesize crystalline MOFs with highly porous topology without introducing extra ligands. For example, Shi et al. synthesized two large pore ZIF compounds with Im organic ligands, by using dibutylformamide (DBF) and dipropylformamide (DPF) solvents as SDAs, respectively²⁹. The obtained ZIFs possess the zeolite CAN and AFI topology with 12-membered rings (MR), and are termed CAN-[Zn(Im)₂] and AFI-[Zn(Im)₂], respectively. AFI-[Zn(Im)₂] displays pore apertures as large as 15.6 Å, which is larger than that of ZIF-4 (2.1 Å, 6 MR) and ZIF-62 (1.4 Å, 6 MR)³⁰. However, CAN-[Zn(Im)₂] and AFI-[Zn(Im)₂] will recrystallize to the dense ZIF-zni phase with 4 MR before melting, similar with ZIF-4^{24,31}. This leads to a reduction in the porosity of the resultant melt-quenched glass, *a_g*-ZIF-zni³².

In this work, meltable MOFs that have highly porous topologies of 12 MR have been synthesised by using both SDAs and mixed ligands of imidazole (Im) and benzimidazole (blm) as shown in Fig. 1. Bulky amide solvents were used as SDAs to create highly porous topologies, without introducing extra ligands. The mixed ligands of Im and blm were used to suppress the recrystallization to dense ZIF-zni phase during the melting process and provide spatial support for porosity preservation. The obtained crystalline frameworks, termed AFI-[Zn(Im)_{1.68}(blm)_{0.32}] and CAN-[Zn(Im)_{1.73}(blm)_{0.27}], possess similar chemical compositions to the prototypical glass former ZIF-62, though they have far greater porosity. The enhanced porosity carries over to the glass upon melt quenching, resulting in high H₂, CO₂, and CH₄ adsorption capacities of *a_g*-AFI-[Zn(Im)_{1.68}(blm)_{0.32}], while decomposition was found during the melt quenching process of CAN-[Zn(Im)_{1.73}(blm)_{0.27}]. Moreover, a free-standing membrane was fabricated with *a_g*-AFI-[Zn(Im)_{1.68}(blm)_{0.32}], featuring record-breaking CH₄/N₂ and CO₂/N₂ separation performance. This porous MOF glass shows promising applications in energy gas purification and CO₂ capture.

Results

Synthesis and Structural Properties of MOF and MOF Glasses AFI-[Zn(Im)₂] and CAN-[Zn(Im)₂] with 12MR pore openings were fabricated by solvothermal synthesis²⁹. The typical ZIF topologies were

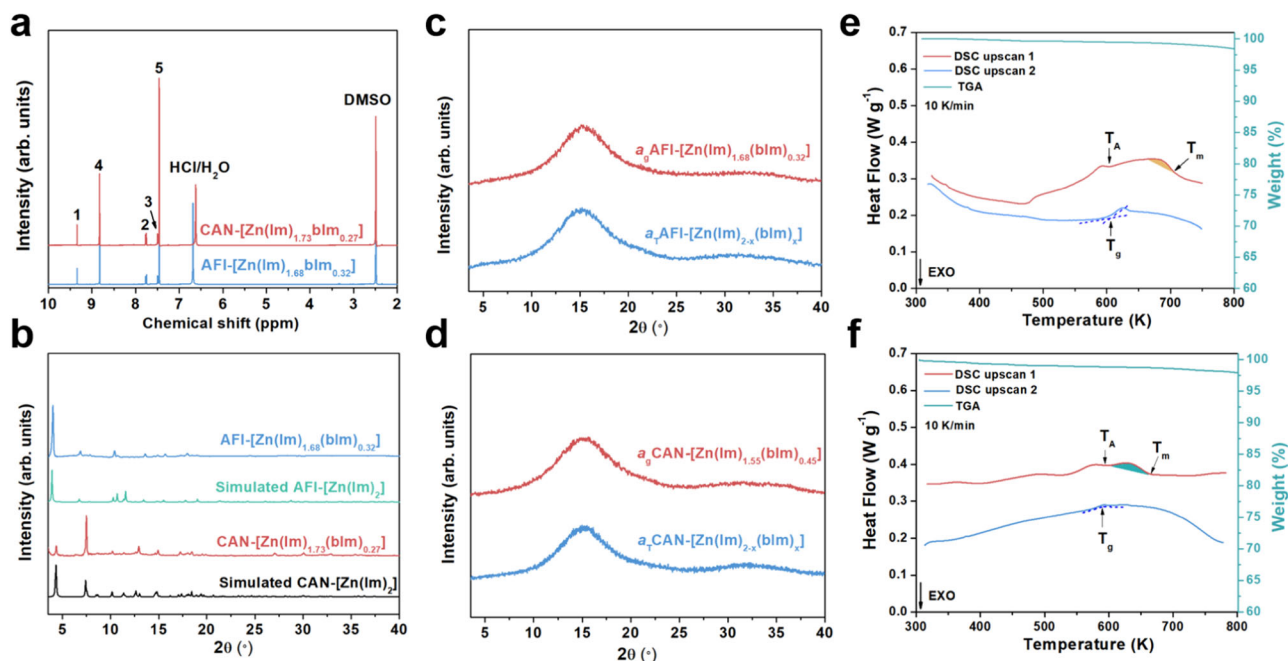


Fig. 2 | Structures and melting behaviours of AFI-[Zn(Im)_{1.68}(bIm)_{0.32}] and CAN-[Zn(Im)_{1.73}(bIm)_{0.27}]. **a** ¹H NMR results of AFI-[Zn(Im)_{1.68}(bIm)_{0.32}] and CAN-[Zn(Im)_{1.73}(bIm)_{0.27}]. **b** XRD patterns of AFI-[Zn(Im)_{1.68}(bIm)_{0.32}], CAN-[Zn(Im)_{1.73}(bIm)_{0.27}], and simulations from the published crystallographic information files (cifs)²⁹. **c**, **d** XRD patterns of amorphous MOFs: (c) α_T AFI-[Zn(Im)_{2-x}(bIm)_x] and α_g AFI-[Zn(Im)_{1.68}(bIm)_{0.32}], and (d) α_T CAN-[Zn(Im)_{2-x}(bIm)_x] and α_g CAN-[Zn(Im)_{1.55}(bIm)_{0.45}].

and α_g CAN-[Zn(Im)_{1.55}(bIm)_{0.45}]. Enthalpic responses and weight loss of (e) AFI-[Zn(Im)_{1.68}(bIm)_{0.32}] and (f) CAN-[Zn(Im)_{1.73}(bIm)_{0.27}] measured by DSC and TGA separately, with a heating rate of 10 K/min. Thermal amorphization (T_A), melting temperature (T_m), and glass transition temperature (T_g) are indicated. The light yellow and blue shaded region highlights the integrated area for the determination of ΔH_m .

listed in Supplementary Fig. 1 and Supplementary Table 1. The porous crystalline structures of AFI-[Zn(Im)₂] and CAN-[Zn(Im)₂] were demonstrated by XRD and N₂ adsorption isotherms (Supplementary Figs. 2 and 3, and Supplementary Table 2). Both AFI-[Zn(Im)₂] and CAN-[Zn(Im)₂] exhibited glass transition temperatures at ca. 590–600 K, and recrystallize to the dense ZIF-zni phase at ca. 800 K, before subsequently melting at 853 K, through partial decomposition of the organic ligand was observed during melting in each case (Supplementary Figs. 4 and 5). Although the chemical composition Zn(Im)₂ is meltable according to previous results, existing Zn(Im)₂ polymorphs all recrystallize to dense phase ZIF-zni before melting^{24,31}. Partial substitution of Im by bIm however has been shown to result in crystalline frameworks of identical topology to ZIF-4, though which melt directly, rather than collapse to a dense framework first¹⁶. The effects of bIm content on the performance of ZIF-62²⁷ and ZIF-8³² have been reported. Henke et al.²⁷ found that the bIm linker partially occupies two of the four available Im salt positions in the ZIF-62 structure. By reducing the ratio of Im and bIm, the T_m of ZIF-62 can be reduced from 703 K to 643 K. The pore size distribution indicates that during the melting and formation of α_g ZIF-62, small cavities (with a diameter of approximately 3.5 Å) completely disappear, but medium-sized cavities (with a diameter of 5–6 Å) are significantly preserved, and even new larger cavities (with a diameter of 8 Å) are formed. In previous studies^{32,33}, bIm was used to replace one of the three methylimidazole (mlm) joints in ZIF-8. As the content of bIm increased, the pore size significantly decreased, the pore size distribution narrowed, and the thermal stability also decreased. To obtain the highly porous structure of MOF glasses, the mixed-linker MOFs were synthesised by controlling ligand ratios of Im and bIm. The NMR results (Fig. 2a, and Supplementary Fig. 8) confirmed the existence of bIm in both frameworks, and the compositions were AFI-[Zn(Im)_{1.68}(bIm)_{0.32}] and CAN-[Zn(Im)_{1.73}(bIm)_{0.27}]. The powder XRD patterns of AFI-[Zn(Im)_{1.68}(bIm)_{0.32}] and CAN-[Zn(Im)_{1.73}(bIm)_{0.27}] were similar to the simulated patterns of AFI-[Zn(Im)₂] and CAN-

[Zn(Im)₂] (Fig. 2b). Pawley refinement (Supplementary Fig. S6 and Table S4) indicated the unit cell dimensions and the space group of AFI-[Zn(Im)_{1.68}(bIm)_{0.32}] and CAN-[Zn(Im)_{1.73}(bIm)_{0.27}] which have little change comparing with AFI-[Zn(Im)₂] and CAN-[Zn(Im)₂]. Although single crystal structures were not obtained, the Pawley refinement results implied that the topology structures of AFI-[Zn(Im)_{1.68}(bIm)_{0.32}] and CAN-[Zn(Im)_{1.73}(bIm)_{0.27}] were similar with that of AFI-[Zn(Im)₂] and CAN-[Zn(Im)₂], respectively. The N₂ adsorption amounts decrease after adding bulky bIm ligand in the framework, for example, the BET decreased from 1155 m²/g to 1073 m²/g for AFI-[Zn(Im)₂] and α_g AFI-[Zn(Im)_{1.68}(bIm)_{0.32}], from 1033 m²/g to 603 m²/g for CAN-[Zn(Im)₂] and α_g CAN-[Zn(Im)_{1.55}(bIm)_{0.45}]. The lower BET surface area of CAN-[Zn(Im)_{1.73}(bIm)_{0.27}] than CAN-[Zn(Im)₂] (Supplementary Fig. 3) is attributed to that the synthesized CAN-[Zn(Im)_{1.73}(bIm)_{0.27}] has low crystallinity, resulting in low porosity of the crystals.

AFI-[Zn(Im)_{1.68}(bIm)_{0.32}] becomes thermally amorphous at 633 K, denoted as α_T , and then melts at 705 K (Fig. 2c, e). The glass obtained from quenching at 735 K possesses an identical composition of α_g AFI-[Zn(Im)_{1.68}(bIm)_{0.32}] confirmed by ¹H NMR (Supplementary Fig. 9a and Fig. 10a–b), and a glass transition temperature of 607 K (Fig. 2e). CAN-[Zn(Im)_{1.73}(bIm)_{0.27}] melts at 670 K, i.e. lower than AFI-[Zn(Im)_{1.68}(bIm)_{0.32}] (Fig. 2d), which was ascribed to the lower bIm content in CAN-[Zn(Im)_{1.73}(bIm)_{0.27}]²⁷. NMR confirms that the α_g CAN glass had composition Zn(Im)_{1.55}(bIm)_{0.45} (Supplementary Fig. 9b and Fig. 10c–d), which indicated some decomposition of the organic linkers upon melt quenching. The α_g CAN-[Zn(Im)_{1.55}(bIm)_{0.45}] possessed a T_g of 591 K that was also lower than α_g AFI-[Zn(Im)_{1.68}(bIm)_{0.32}]. The synthesized CAN-[Zn(Im)_{1.73}(bIm)_{0.27}] has low crystallinity and parasitic non-porous phase, resulting in low porosity of the crystals. The melting points of both AFI-[Zn(Im)_{1.68}(bIm)_{0.32}] and CAN-[Zn(Im)_{1.73}(bIm)_{0.27}] were similar to that of the prototypical ZIF-62 (T_m = 710 K). In addition, the SEM images displayed aggregate needle-like morphologies for both AFI-[Zn(Im)_{1.68}(bIm)_{0.32}] and CAN-

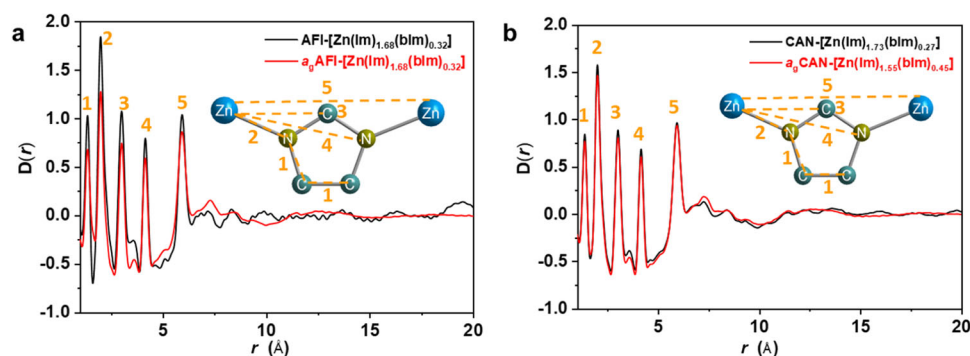


Fig. 3 | Amorphous structures of the melt quenched glasses. Pair distribution functions, $D(r)$, of (a) AFI-[Zn(Im)_{1.68}(blm)_{0.32}] and a_g AFI-[Zn(Im)_{1.68}(blm)_{0.32}], and (b) CAN-[Zn(Im)_{1.73}(blm)_{0.27}] and a_g CAN-[Zn(Im)_{1.55}(blm)_{0.45}], and the inserts assign

PDF peaks to the main short atom-atom distances in the structure. Zn, blue; C, dark green; N dark yellow; H, omitted.

Table 1 | Gas capacities and pore volumes obtained by 195 K CO₂ isotherms

Sample	Method	V_{ads} (cm ³ /g)	n_{ads} (mmol/g)	S_{BET} (m ² /g)	V_{pore} (cm ³ /g)	Ref.
a_g ZIF-62	195 K CO ₂	74.8	3.34	200	0.12	34
a_g TIF-4	195 K CO ₂	76.2	3.4	204	0.12	34
a_g ZIF-4	195 K CO ₂	63.6	2.84	187	0.10	34
a_g ZIF-67-mim _{0.18} im _{0.68} blm _{0.14}	195 K CO ₂	118.5	5.3	316	0.18	35
a_g ZIF-8-mim _{0.15} im _{0.74} blm _{0.11}	195 K CO ₂	125.0	5.6	403	0.20	35
a_g CAN-[Zn(Im) _{1.55} (blm) _{0.45}]	195 K CO ₂	83.5	3.72	241	0.14	This work
a_g AFI-[Zn(Im) _{1.68} (blm) _{0.32}]	195 K CO ₂	91.0	4.06	285	0.16	This work

[Zn(Im)_{1.73}(blm)_{0.27}], and bulk boundary-free morphology for corresponding MOF glasses (Supplementary Fig. 11). Here, we observed solid-state thermal amorphization, prior to melting for both AFI-[Zn(Im)_{1.68}(blm)_{0.32}] and CAN-[Zn(Im)_{1.73}(blm)_{0.27}]. Importantly, we did not observe recrystallization to ZIF-zni phase for either AFI-[Zn(Im)_{1.68}(blm)_{0.32}] or CAN-[Zn(Im)_{1.73}(blm)_{0.27}] (Fig. 2c, d). Moreover, DSC testing of AFI-[Zn(Im)_{1.68}(blm)_{0.32}] and CAN-[Zn(Im)_{1.73}(blm)_{0.32}] were evaluated at varying heating rates (10 K, 20 K, and 30 K), the T_g , T_m , and T_d were marked in Supplementary Figs. 12, 13 and Supplementary Table 5. The DSC results demonstrated that both AFI-[Zn(Im)_{1.68}(blm)_{0.32}] and CAN-[Zn(Im)_{1.73}(blm)_{0.32}] exhibited melting characteristics.

The amorphous nature of the melt quenched a_g AFI-[Zn(Im)_{1.68}(blm)_{0.32}] and a_g CAN-[Zn(Im)_{1.55}(blm)_{0.45}] glasses was confirmed by the structure factors $S(Q)$ obtained through room-temperature X-ray total scattering (Supplementary Fig. 15). Subsequent Fourier transform yielded the pair distribution functions (PDFs) $D(r)$ (Fig. 3 and Supplementary Fig. 16). The PDFs of both a_g AFI-[Zn(Im)_{1.68}(blm)_{0.32}] and a_g CAN-[Zn(Im)_{1.55}(blm)_{0.45}] were essentially featureless above 8 Å, indicating long-range disorder. Below 8 Å, PDF traces for both crystals and glasses are virtually identical, with only small differences in peak intensities.

Mechanical stability and porosity

The processability of MOF glasses is an important advantage compared with MOF crystals, allowing customization for various applications. Assessing the material's mechanical properties is crucial for determining its ability to withstand industrial conditions²⁰. The mechanical properties of a_g AFI-[Zn(Im)_{1.68}(blm)_{0.32}] and a_g CAN-[Zn(Im)_{1.55}(blm)_{0.45}] were measured by using nano-indentation (Supplementary Fig. 17). The Young's modulus (E) was 6.58 ± 0.10 GPa and 6.63 ± 0.05 GPa, and the hardness (H) was 0.670 ± 0.013 GPa and 0.656 ± 0.013 GPa for a_g AFI-[Zn(Im)_{1.68}(blm)_{0.32}] and a_g CAN-[Zn(Im)_{1.55}(blm)_{0.45}], respectively, which are similar to those of a_g ZIF-62 ($E = 6.58 \pm 0.02$ GPa, $H = 0.656 \pm 0.005$ GPa).

Due to the relatively low BET surface area of CAN-[Zn(Im)_{1.73}(blm)_{0.27}] and the slight decomposition during melting, the investigation of gas separation performance was focused on AFI-[Zn(Im)_{1.68}(blm)_{0.32}] and a_g AFI-[Zn(Im)_{1.68}(blm)_{0.32}]. The gas isotherms of AFI-[Zn(Im)_{1.68}(blm)_{0.32}] and a_g AFI-[Zn(Im)_{1.68}(blm)_{0.32}] were tested by H₂, CO₂, N₂, and CH₄ (Fig. 4a–d). As expected, crystalline AFI-[Zn(Im)_{1.68}(blm)_{0.32}] with open nano-pores was porous to a range of small gases, and hysteresis in the desorption branches was observed for H₂ at 77 K and CH₄ at 273 K. Glassy a_g AFI-[Zn(Im)_{1.68}(blm)_{0.32}] showed lower gas uptake than the crystal for all tested gases. The partial collapse of open pores in crystalline MOFs during melt-quenching results in a decrease in porosity in MOF glasses^{21,34}. No hysteresis was observed for N₂, indicating that N₂ molecules desorb rapidly due to their weaker interactions with a_g AFI-[Zn(Im)_{1.68}(blm)_{0.32}], resulting in the absence of a hysteresis effect during N₂ desorption, similar case has been reported for a_g ZIF-8-mim_{0.15}im_{0.74}blm_{0.11}³⁵. Moreover, the first step of pore collapse also occurred at the point of thermal amorphization, as significant decrease in N₂ uptake was observed for a_T AFI-[Zn(Im)_{1.68}(blm)_{0.32}] (Supplementary Fig. 18). The porosity and pore size distribution of a_g AFI-[Zn(Im)_{1.68}(blm)_{0.32}] and a_g CAN-[Zn(Im)_{1.55}(blm)_{0.45}] were measured by CO₂ gas adsorption at 195 K (Fig. 4e and Supplementary Fig. 19), and the comparison was made with other MOF glasses in Table 1^{34,35}. As shown in Fig. 4f, the high-pressure CO₂ gas adsorption at 273 K and 3000 kPa demonstrated the enhanced CO₂ adsorption capacity of a_g AFI-[Zn(Im)_{1.68}(blm)_{0.32}]. As shown in Supplementary Fig. 20, following the melting and quenching process, both pre-pressing and post-pressing a_g AFI-[Zn(Im)_{1.68}(blm)_{0.32}] exhibited an almost identical CO₂ adsorption capacity. Although mechanical pressure slightly damages the porous structure of AFI-[Zn(Im)_{1.68}(blm)_{0.32}] crystals, it has almost no effects on the adsorption capacity of the melt-quenching glass.

Furthermore, as shown in Supplementary Fig. S21, the a_g AFI-[Zn(Im)_{1.68}(blm)_{0.32}] was heated above the glass transition temperature ($T_g = 673$ K) for 2 h, resulting in the formation of the a_g AFI-

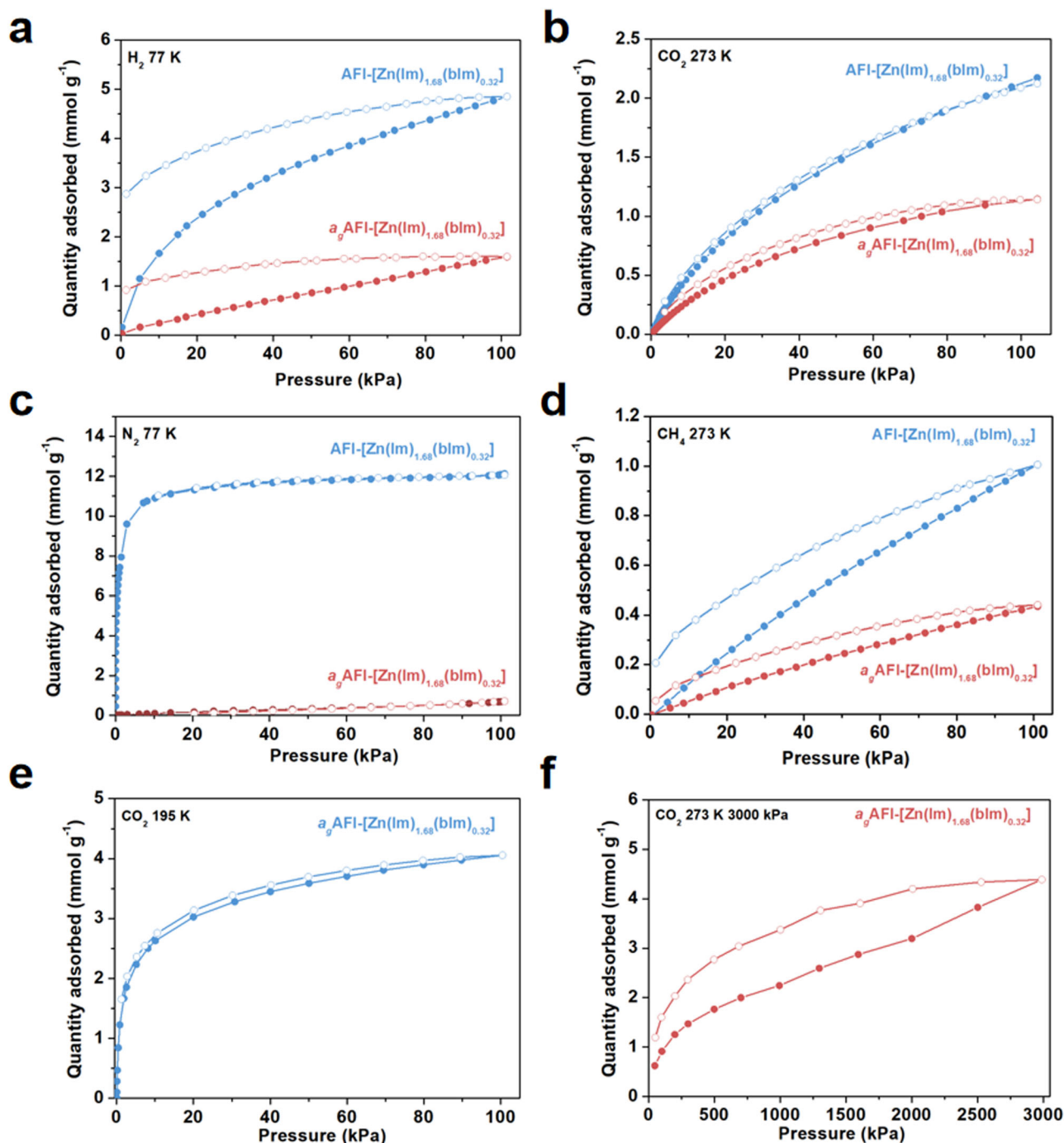


Fig. 4 | Gas isotherms of $\text{AFI}[\text{Zn}(\text{Im})_{1.68}(\text{bIm})_{0.32}]$ and $a_g\text{AFI}[\text{Zn}(\text{Im})_{1.68}(\text{bIm})_{0.32}]$. Results were tested by (a) H_2 at 77 K, (b) CO_2 at 273 K, (c) N_2 at 77 K, (d) CH_4 at 273 K, (e) CO_2 at 195 K, and (f) CO_2 at 273 K up to 3000 kPa. The filled and hollow circles represent adsorption and desorption, respectively.

$[\text{Zn}(\text{Im})_{1.68}(\text{bIm})_{0.32}]_{\text{T}}$. The slight reduction in CO_2 adsorption capacity is indicative of the sufficient collapse of the pores in the $a_g\text{AFI}[\text{Zn}(\text{Im})_{1.68}(\text{bIm})_{0.32}]$ following the initial adequate heat treatment.

Remarkably, $a_g\text{AFI}[\text{Zn}(\text{Im})_{1.68}(\text{bIm})_{0.32}]$ displayed significantly higher gas adsorption quantity for H_2 , CO_2 , and CH_4 than other MOF glasses and CGCs such as $a_g\text{ZIF-62}$ and $(\text{MIL-118})_{0.5}(a_g\text{ZIF-62})_{0.5}$ ³⁶ (Supplementary Table 6). For example, $a_g\text{AFI}[\text{Zn}(\text{Im})_{1.68}(\text{bIm})_{0.32}]$ absorbs 1.56 mmol g^{-1} H_2 at 77 K and a pressure of 100 kPa, and the adsorption quantity of $a_g\text{ZIF-62}$ and $a_g[(\text{ZIF-8})_{0.2}(\text{ZIF-62})_{0.8}]$ is 0.42 and 1.24 mmol g^{-1} , respectively. The low density of $a_g\text{AFI}[\text{Zn}(\text{Im})_{1.68}(\text{bIm})_{0.32}]$ when compared with $a_g\text{ZIF-62}$ is consistent with the higher porosity. Moreover, the low density of $a_g\text{AFI}[\text{Zn}(\text{Im})_{1.68}(\text{bIm})_{0.32}]$

possibly indicates the presence of a multitude of blocked or inaccessible pores. During the transformation from crystal to glass, the collapse of the pore structure results in the formation of varying degrees of pore channels. Furthermore, it is unavoidable that some pores will become obstructed, preventing He from entering and resulting in a reduction in the He pycnometric density. The $a_g\text{AFI}[\text{Zn}(\text{Im})_{1.68}(\text{bIm})_{0.32}]$ demonstrated a similar H_2 adsorption quantity as $(\text{MIL-118})_{0.5}(a_g\text{ZIF-62})_{0.5}$ and $(\text{UL-MOF-1})_{0.5}(a_g\text{ZIF-62})_{0.5}$, both of which contain 50 wt% porous MOF crystals³⁷. Although $a_g\text{AFI}[\text{Zn}(\text{Im})_{1.68}(\text{bIm})_{0.32}]$ and $a_g\text{ZIF-62}$ showed high similarity in chemical composition, glass transition temperature, PDFs, and mechanical properties, these two MOF glasses showed significant differences in density and gas uptake, which is believed to be

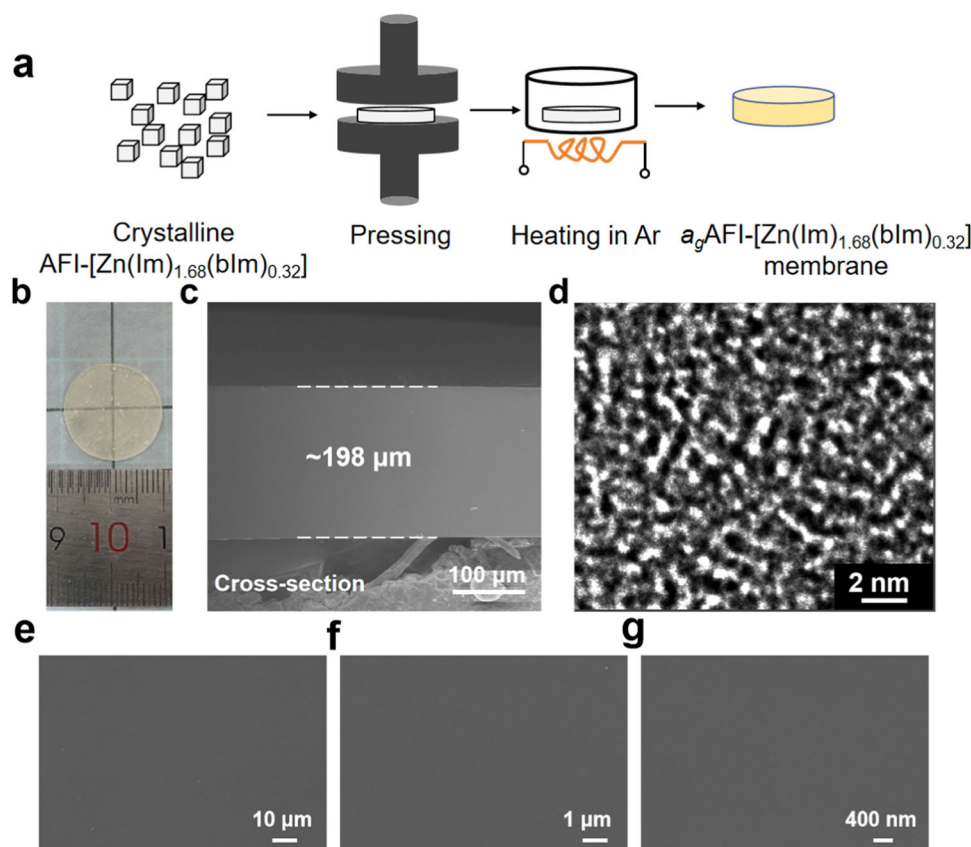


Fig. 5 | Fabrication process and morphologies of the $a_g\text{AFI}[\text{Zn}(\text{Im})_{1.68}(\text{bIm})_{0.32}]$ membrane. **a** Fabrication process. **b** Photograph. **c** Cross-section SEM image. **d** HR-TEM dark field image. **e–g** Surface SEM images at different magnification.

attributed to the difference in topology of the crystalline precursors. It was the avoidance of recrystallization (to ZIF-zni) that we believe is responsible for imparting a higher porosity to the resultant glasses.

Unlike the conventional inorganic glasses whose microporosity is generally produced by leaching, microporosity is an intrinsic property of MOF glasses²¹. The molecular linkers of MOFs, especially the bulkier linkers, play an important role in suppressing the formation of densely packed structures upon melting^{32,38}. Moreover, the large solvent molecules used as SDAs in the synthesis can create highly porous topologies in crystalline MOFs without introducing extra ligands that possibly decrease the decomposition temperature or increase the melting point of the resultant MOF crystals. Thus, the porosity of MOF glasses can be significantly improved by increasing the porosity of crystalline precursors through using both SDAs and mixed ligands of Im and bIm in the synthesis process.

Membrane fabrication and separation performance

The increased material processability, combined with pore structure, makes the glassy MOFs prepared in this work attractive for applications such as separation membranes, adsorbents, and beyond. To illustrate this, the membranes were successfully prepared by easily processing the $a_g\text{AFI}[\text{Zn}(\text{Im})_{1.68}(\text{bIm})_{0.32}]$ into mechanically durable pallets with a diameter of 2 cm (Fig. 5a) through their molten phase. The obtained membrane is transparent and self-supported (Fig. 5b). The cross-section (Fig. 5c) and surface (Fig. 5e–g) of the membrane were examined by SEM, showing a smooth and dense membrane structure without visible voids or cracks. Overall, the membrane had a high degree of integrity and uniformity. The thickness of the membranes, ranged from ~100 μm, ~200 μm, and ~300 μm (Supplementary Fig. 22) and could be controlled by varying the mass of $\text{AFI}[\text{Zn}(\text{Im})_{1.68}(\text{bIm})_{0.32}]$ used in the pressing before melting. Although the

membrane displayed a dense morphology in the SEM images, the microstructure nature of $a_g\text{AFI}[\text{Zn}(\text{Im})_{1.68}(\text{bIm})_{0.32}]$ could be demonstrated from FIB lamella under the HR-TEM dark field mode (Fig. 5d). The microstructure of $a_g\text{AFI}[\text{Zn}(\text{Im})_{1.68}(\text{bIm})_{0.32}]$ was observed to exhibit slight looseness than that of $a_g\text{ZIF-62}$ (Supplementary Fig. 23), and the packing density was decreased when analysed using HR-TEM.

The gas separation performance of the self-supported $a_g\text{AFI}[\text{Zn}(\text{Im})_{1.68}(\text{bIm})_{0.32}]$ membrane was measured by using CO_2 , N_2 , and CH_4 pure gases, and mixed gases of CO_2/N_2 (50/50) and CH_4/N_2 (50/50). The $a_g\text{AFI}[\text{Zn}(\text{Im})_{1.68}(\text{bIm})_{0.32}]$ membrane with thickness of ~200 μm showed high gas permeance tested by using pure gases, for example CO_2 permeance of 5.1×10^4 GPU, and CH_4 permeance of 1.6×10^4 GPU (Fig. 6a). The ideal gas selectivities, determined from the pure gas measurements, gave CO_2/N_2 selectivity of 19.5 and CH_4/N_2 selectivity of 6.1, were much higher than the corresponding Knudsen selectivities (Fig. 6b). The separation performance of the $a_g\text{AFI}[\text{Zn}(\text{Im})_{1.68}(\text{bIm})_{0.32}]$ membrane with thickness of ~200 μm obtained using mixed gases (Fig. 6a, b) showed CO_2 permeance of 3.7×10^4 GPU with CO_2/N_2 selectivity of 14.8 and CH_4 permeance of 1.3×10^4 GPU with CH_4/N_2 selectivity of 5.2. Furthermore, as shown in Supplementary Fig. 24, the adsorption curves of CH_4 and N_2 demonstrate that $a_g\text{AFI}[\text{Zn}(\text{Im})_{1.68}(\text{bIm})_{0.32}]$ exhibited a higher adsorption capacity for CH_4 than for N_2 . Hence, the gas separation performance of the $a_g\text{AFI}[\text{Zn}(\text{Im})_{1.68}(\text{bIm})_{0.32}]$ membrane for CH_4/N_2 was primarily determined by adsorption selectivity, as evidenced by the results of gas adsorption and permeation testing.

Compared with the results obtained by pure gases, the separation performance tested by mixed gases was slightly reduced, which was mainly caused by the competitive adsorption effect between mixed gases. The effects of membrane thickness for mixed gas were also investigated (Supplementary Table 3). With increasing thickness from

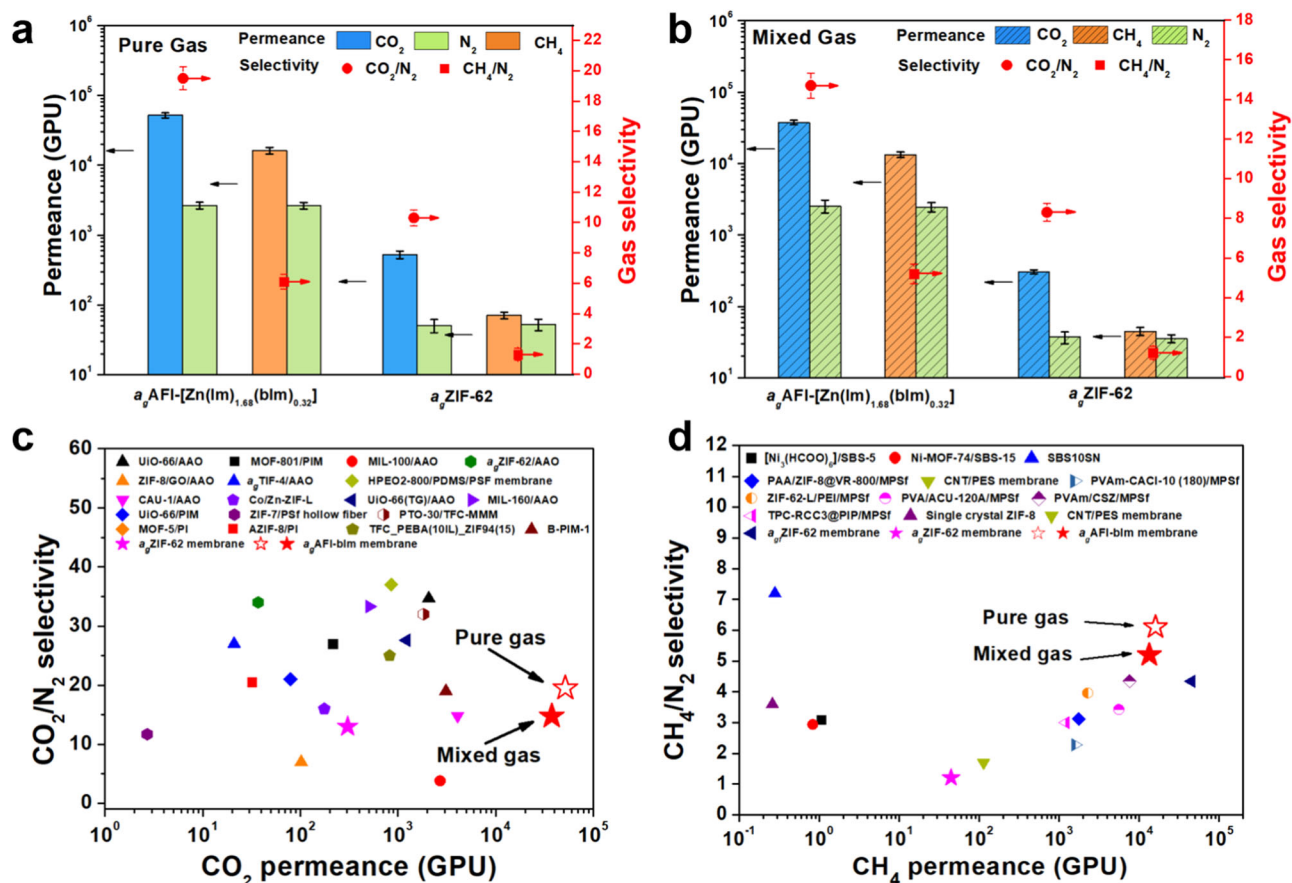


Fig. 6 | Gas separation performance of the *a_g*AFI-[Zn(Im)]_{1.68}(blM)_{0.32} membrane tested by both pure and mixed gases at 298 K and 50 kPa. a, b Gas permeance and gas selectivity for pure gas and mixed gas. Comparison of (c) CO₂/N₂

and (d) CH₄/N₂ separation performance of the *a_g*AFI-[Zn(Im)]_{1.68}(blM)_{0.32} membrane with other reported high-performance membranes. Error bars represent standard deviations.

~100 to ~300 μm, the CH₄ permeance decreases from 1.55×10^4 GPU to 1.04×10^4 GPU, and the CH₄/N₂ selectivity increases from 3.8 to 6.4. The *a_g*AFI-[Zn(Im)]_{1.68}(blM)_{0.32} membrane had the developed pore continuity and improved porosity, which effectively reduced the path resistance of gas molecules through the membrane. The considerable enhancement in permeance suggested that the primary gas transport mechanism may be direct diffusion through the pores, with the separation mechanism being dominated by the synergistic interaction forces. The pycnometric density test indicated the formation of a series of dead-end pores in the *a_g*AFI-[Zn(Im)]_{1.68}(blM)_{0.32} membrane, which reduced the actual separation thickness of the glass membrane and enhanced the gas permeance. In addition, the effect of long-term stability was investigated by using mixed gases (Supplementary Fig. 25). Both CH₄ permeance and CH₄/N₂ selectivity remain stable within 100 h, suggesting good stability of the *a_g*AFI-[Zn(Im)]_{1.68}(blM)_{0.32} membrane.

For comparison, a self-supported *a_g*ZIF-62 membrane (~200 μm) was fabricated and tested by using mixed gases. The *a_g*AFI-[Zn(Im)]_{1.68}(blM)_{0.32} membrane (~200 μm) possessed much higher gas permeance and selectivity than the *a_g*ZIF-62 membrane for both CO₂/N₂ and CH₄/N₂ gas pairs (Fig. 6a, b). The higher gas selectivity of *a_g*AFI-[Zn(Im)]_{1.68}(blM)_{0.32} membrane than that of the *a_g*ZIF-62 membrane (14.8 vs 8.3 for CO₂/N₂ and 5.2 vs 1.2 for CH₄/N₂) was reasonably attributed to the higher porosity of *a_g*AFI-[Zn(Im)]_{1.68}(blM)_{0.32} that leads to higher gas uptake as well as higher CO₂/N₂ and CH₄/N₂ solubility selectivity (Supplementary Tables 8, 9). The difference in gas permeance between *a_g*AFI-[Zn(Im)]_{1.68}(blM)_{0.32} and *a_g*ZIF-62 membranes (3.7×10^4 vs 305 GPU for CO₂/N₂ and 1.3×10^4 vs 45 GPU for CH₄/N₂) seemed not to be simply attributable to the different porosities of

the glasses, considering the lesser difference in CH₄ uptake between *a_g*AFI-[Zn(Im)]_{1.68}(blM)_{0.32} and *a_g*ZIF-62 powders (0.45 vs 0.17 mmol g⁻¹) (Supplementary Table 1). The possible reason is that, besides of the higher porosity, the *a_g*AFI-[Zn(Im)]_{1.68}(blM)_{0.32} membrane has a larger pore size and better pore connectivity, which results in higher diffusivity coefficient of gas molecules.

The gas separation performance of *a_g*AFI-[Zn(Im)]_{1.68}(blM)_{0.32} membrane was compared with the reported high-performance microporous membranes including crystalline MOFs, *a_g*ZIF-62, 2D materials, porous organic cages, and mixed matrix membranes (MMMs) as shown in Fig. 6c, d and Supplementary Tables 10, 11. The *a_g*AFI-[Zn(Im)]_{1.68}(blM)_{0.32} membrane has the highest CO₂ permeance and the next-highest CH₄ permeance and the gas selectivity is comparable with the reported membranes. The high gas permeance of the *a_g*AFI-[Zn(Im)]_{1.68}(blM)_{0.32} membrane is favourable to reduce the required membrane area of the separation processes, which could largely decrease the operating cost of gas separation processes^{39–41}. The *a_g*AFI-[Zn(Im)]_{1.68}(blM)_{0.32} membrane is suitable for assembling into disc type membrane modules. Although the packing density of disc type membrane modules is typically lower than that of spiral and hollow fibre modules, they can effectively inhibit concentration polarization of permeation gas, which is conducive to high-permeance membrane assembly. Disc type membrane modules can be designed to effectively handle gases in confined spaces, such as small pollution sources or compact waste gas treatment devices.

Comparing with the reported three types of *a_g*ZIF-62 membranes^{22,26}, the *a_g*AFI-[Zn(Im)]_{1.68}(blM)_{0.32} membrane shows significant advances in gas separation performance. Wang²² et al. prepared an *a_g*ZIF-62 thin-film composite membrane by using porous

anodic aluminium oxide (AAO) as the support, which is a practical strategy to reduce the thickness of the separation layer and increase the gas permeance. However, the $a_g\text{ZIF-62/AAO}$ composite membrane displays only a CO_2 permeance of 37.6 GPU with a CO_2/N_2 selectivity of 34, due to the penetration of $a_g\text{ZIF-62}$ into the pores of AAO. The support layer on $a_g\text{ZIF-62/AAO}$ surface can penetrate and increase the actual thickness of the composite membrane, thereby enhancing the CO_2/N_2 separation performance. Consequently, the CO_2/N_2 selectivity of the $a_g\text{ZIF-62/AAO}$ membrane as in the literature is higher than that of the self-supported $a_g\text{ZIF-62}$ in this work. Knebel et al.⁴² produced $a_g\text{ZIF-62}$ with varying degrees of pore collapse by modifying the process parameters of melt-quenching. This strategy prevents the pore collapse during melt-quenching, and results in a CO_2 diffusivity of $3.53 \times 10^{-8} \text{ cm}^2/\text{s}$ measured by an infrared microscope that cannot be compared directly to the results measured by gas permeation used in this work. In our previous work²⁶, a ZIF-62 glass foam, i.e., $a_g\text{ZIF-62}$, was developed by melting ZIF-62 with polyethyleneamine as the pore-forming agent, and the porosity of the formed glass was largely improved through the decomposition of polyethyleneamine. The $a_g\text{ZIF-62}$ membrane displays a higher CH_4 permeance of 4.85×10^4 GPU and a lower CH_4/N_2 selectivity of 4.08 than those of $a_g\text{AFI-[Zn(Im)}_{1.68}(\text{blm})_{0.32}]$ in this work. The higher gas selectivity of $a_g\text{AFI-[Zn(Im)}_{1.68}(\text{blm})_{0.32}]$ than that of $a_g\text{ZIF-62}$ demonstrates that the strategy of designing melttable crystalline MOFs with large-pore topology in this work has advantages in achieving precise manipulation of porosity in MOF glass at the Ångström scale.

Discussion

Melttable crystalline MOFs $\text{AFI-[Zn(Im)}_{1.68}(\text{blm})_{0.32}]$ and $\text{CAN-[Zn(Im)}_{1.73}(\text{blm})_{0.27}]$ were synthesised by using bulky amide solvents as SDAs and mixed ligands of Im and blm. $\text{AFI-[Zn(Im)}_{1.68}(\text{blm})_{0.32}]$ and $\text{CAN-[Zn(Im)}_{1.73}(\text{blm})_{0.27}]$ exhibit large pore topologies, and no recrystallization to ZIF-zni was observed before melting, which partially preserves the highly porous topology in the resultant melt-quenched glasses. The gas adsorption capacity of $a_g\text{AFI-[Zn(Im)}_{1.68}(\text{blm})_{0.32}]$ is higher than reported MOF glasses, for example, H_2 and CO_2 adsorption capacities of 1.56 mmol g^{-1} and 1.12 mmol g^{-1} , respectively. Self-supported $a_g\text{AFI-[Zn(Im)}_{1.68}(\text{blm})_{0.32}]$ membranes were easily fabricated due to its high processability and they show record-breaking gas separation performance tested by mixed gases, for example, CO_2 permeance of 3.7×10^4 GPU with CO_2/N_2 selectivity of 14.8, and CH_4 permeance of 1.3×10^4 GPU with CH_4/N_2 selectivity of 5.2. The porosity and gas separation performance of MOF glasses have been significantly improved by designing highly porous topologies into the crystalline precursors, which may lay a foundation for the application of MOF glass in low-cost gas separation process.

Methods

Synthesis of MOFs

$\text{AFI-[Zn(Im)}_{1.68}(\text{blm})_{0.32}]$ and $\text{CAN-[Zn(Im)}_{1.73}(\text{blm})_{0.27}]$ were synthesized according to the previous report with some modification²⁹. Specifically, $\text{AFI-[Zn(Im)}_{1.68}(\text{blm})_{0.32}]$: Zinc acetate (286 mg, 1.56 mmol) was dissolved in N,N-dipropylformamide (DPF, 5 mL) in a 20 mL glass vial. Imidazole (309 mg, 4.55 mmol) and benzimidazole (76.6 mg, 0.65 mmol) were dissolved in DPF (10 mL) in a separate glass vial. The solutions were mixed and placed in an oven at 328 K for 7 days. After cooling to room temperature, the product was replaced with dry acetone three times a day for a total of three days. $\text{CAN-[Zn(Im)}_{1.73}(\text{blm})_{0.27}]$ was synthesized by replacing DPF with dibutylformamide (DBF) as the solvent, and other conditions were as same as that for $\text{AFI-[Zn(Im)}_{1.68}(\text{blm})_{0.32}]$. According to the reported literature, ZIF-62 was synthesized by solvothermal method²⁰. Zinc nitrate hexahydrate ($\text{Zn(NO}_3)_2 \cdot 6\text{H}_2\text{O}$, 1.65 g, 5.54 mmol), Im (8.91 g, 131 mmol), and blm (1.55 g, 131 mmol) were dissolved in 75 mL N,N-

dimethylformamide (DMF) in a glass jar. The mixture was statically reacted in an oven at 403 K for 48 h. The product was then cooled to room temperature, and washed several times with dichloromethane dried in an oven at 313 K.

Characterization

Powder X-ray diffraction (XRD) analysis ($2\theta = 5^\circ$ to 40°) was collected at room temperature with a Bruker-AXS D8 diffractometer with $\text{Cu K}\alpha$ ($\lambda = 1.540598 \text{ \AA}$) radiation detector. The Pawley refinement was performed using TOPAS (Version 3) program. X-ray total scattering data were collected on the I15-1 beamline at the Diamond Light Source (UK) with an X-ray wavelength of $0.161669 \text{ \AA}$ (76.7 keV). The BL14B1 beamline at the Shanghai Synchrotron Radiation Facility (SSRF, China) was used by an X-ray wavelength of $0.6889 \text{ \AA}$ (18 keV). A series of PDFs pre-experiments were carried out using SSRF in the initial and validation stages of the investigation, with the main purpose of verifying the feasibility of the experimental design and confirming the structure of the samples, while the data shown in this manuscript were obtained from the I15-1 beamline at the Diamond Light Source. Samples were loaded into borosilicate glass capillaries of 1.17 mm (inner) diameter. Data from the samples, empty instrument and empty capillary were collected and processed using the GudrunX program. The surface morphologies of samples were observed by using a scanning electron microscope (SEM), FEI Nova Nano 450. The density of the glasses was measured by pycnometry on a Micromeritics AccuPyc 1340 instrument with He gas. Gas sorption isotherms were conducted on a Micromeritics ASAP 2020 instrument with around 200 mg samples. High-pressure CO_2 adsorption tests were conducted on ZIF glasses using a high-pressure gas adsorption analyser (BSD-PH, Bei Shi De) at 273 K, with a pressure range of 0–3000 kPa. Prior to the adsorption/desorption testing, all samples were degassed under vacuum at 353 K for 6 h. The saturation pressure of 191 kPa was used for specific surface area calculation from CO_2 sorption isotherms recorded at 195 K^{34,35}. The thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC) analysis were conducted using TA Q600 instrument in an argon atmosphere at a heating rate of 10 K/min, 20 K/min, and 30 K/min. The solution ^1H nuclear magnetic resonance (^1H NMR) spectra were recorded on a Bruker Avance III 500 MHz spectrometer at 293 K. The digested samples were prepared by mixing a mixture of DCl (35%)/ D_2O (0.1 mL) and DMSO-d_6 (0.5 mL) with MOFs (about 6 mg). The chemical shift refers to the DMSO-d_6 solvent signal and the spectrum is processed using the MestreNova Suite. The Young's modulus (E) and hardness (H) of MOF glasses were studied via depth-sensing indentation testing with a nanoindenter (G200, Agilent Inc.), equipped with a three-sided Berkovich diamond tip (Synton-MDP Inc.). The MOF glasses were first mounted in a thermosetting epoxy resin (Araldite CY212, Agar Scientific Ltd.) and polished to optical finish using a water-based diamond suspension. The values of E were estimated for a theoretical Poisson ratio, ν , of 0.20. Prior to the first experiments, the area function of the Berkovich indenter tip as well as the instruments frame compliance were calibrated on a fused silica reference glass sample of known elastic properties (Corning Code 7980, Corning Inc.).

Preparation of MOF glass membranes

The glassy MOF membranes were fabricated by pressing and melt-quenching. Crystalline $\text{AFI-[Zn(Im)}_{1.68}(\text{blm})_{0.32}]$ powders were pressed at 10 MPa pressure to form a pellet at first. The crystalline pellet was then put into a tube furnace under Ar atmosphere, heated to 705 K with a heating rate of 10 K/min, and kept at 705 K for 30 min. After cooling down to the room temperature, the $a_g\text{AFI-[Zn(Im)}_{1.68}(\text{blm})_{0.32}]$ membrane was obtained. The $a_g\text{ZIF-62}$ membranes were prepared by using a maximum temperature of 713 K, and other conditions were the same as that for the $a_g\text{AFI-[Zn(Im)}_{1.68}(\text{blm})_{0.32}]$ membranes.

Characterization of MOF glass membranes

The surface and cross-section morphology of membranes were observed by using scanning electron microscope-dual beam focused ion beam (SEM-FIB, CrossBeam 550, Zeiss). The membrane microstructure was observed by high resolution transmission electron microscope (HR-TEM, JEM-F200, Jeol) with a Gatan CMOS CCD camera. The instrument was operated at 200 kV. The TEM samples were prepared by ultrathin-sectioning by the FIB equipped with a ZEISS CrossBeam 550 instrument. The samples were sectioned at 30 kV ion gun voltage, rough sectioning at 30 nA current, and polished at 1 nA.

Gas permeation measurements

The mixed gas separation performance of membranes was evaluated by constant pressure-variable volume method. The gas selectivity (α) was the ratio between two components P_i and P_j . For the mixed gas testing, a gas chromatograph (Agilent 8890) was used to measure the concentration of mixed gases on the permeation side driven by the sweep gas.

The gas permeance of glass membranes were measured by using constant pressure-variable volume method with Eq. (1):

$$P_i = \frac{Q_i}{S \Delta P_i} \quad (1)$$

where P_i is the gas permeance of i -component (GPU, 1 GPU = 10^{-6} cm³ (STP) cm⁻² s⁻¹ cm-Hg⁻¹), Q_i is the molar flow rate of i -component (cm³ (STP)/s), S is the membrane area (cm²), and ΔP_i is the pressure difference across the membrane (Pa).

The solubility coefficient (S) and diffusion coefficient (D) of membranes were measured at 50 kPa and 298 K via the constant volume-variable pressure method with pure gas (time-lag method) on the Labthink instrument, G2/110-A. The initial step involved the application of a vacuum treatment to both the feed side and the permeation side. Subsequently, the cavity on the permeation side was sealed, the feed gas side was filled with test gas at a specified pressure, and it was ensured that a pressure differential of 50 kPa was established on both sides of the glass membrane. Subsequently, the gas will permeate from the feed side to the permeation side under the pressure gradient. The permeance, solubility coefficient, and diffusion coefficient of the membrane can be calculated by processing the pressure data from the permeable side using the software.

The diffusion coefficient (D , cm²/s) was calculated by time-lag method (2):

$$D = \frac{l^2}{6\theta} \quad (2)$$

where l was the membrane thickness and θ was the time lag (s) before the steady straight-line relationship was calculated by the downstream pressure versus time plot.

The solubility coefficient S (cm³(STP)·cm⁻³·cmHg⁻¹) was given by Eq. (3):

$$S = \frac{P}{D} \quad (3)$$

Data availability

The authors declare that all data supporting the findings of this study are available within the paper and Supplementary Information files. Source data are provided with this paper.

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Author contributions

S.L., Z.Q. and T.D.B. designed the project. MOFs were synthesized by S.L. and J.H. Membrane preparation and characterization were carried out by C.M., A.C., J.D., Z.Q. and C.Z. PDF measurements were carried out by S.L., P.A.C., D.S.K. and D.A.K. Gas isotherm measurements were carried out by S.L. S.Y. and Y.L. S.L. and Z.Q. wrote the manuscript with T.D.B. with input from all authors. All authors performed certain experiments and discussed and revised the paper.

Competing interests

The authors declare no competing interests.

Additional information

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